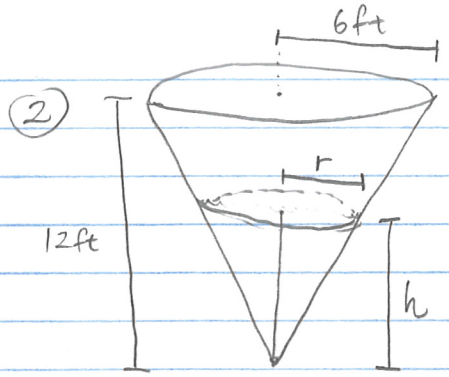
(2)

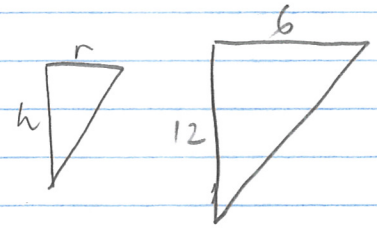


Volume of cone

$$V = \frac{1}{3}\pi r^2 h$$

$$V = \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h$$

$$V = \frac{\pi}{12}h^3$$



$$\frac{r}{h} = \frac{6}{12}$$

$$r = \frac{h}{2}$$

Q why solve for r? instead of h?

deriv wrt t

$$\frac{dV}{dt} = \frac{\pi}{12} \cdot 3h^2 \cdot \frac{dh}{dt}$$

Given: $\frac{dV}{dt} = 8 \quad h = 4$

Find $\frac{dh}{dt}$

$$8 = \frac{\pi}{12} \cdot 3(4)^2 \cdot \frac{dh}{dt}$$

$$\frac{8 \cdot 12}{\pi \cdot 3 \cdot 4^2} = \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{2}{\pi} \approx 0.637$$

The drink's height is rising at a rate of 0.637 ft/min.

(Think: near the end, it takes more time for the drink level to rise)